Another variant of the hot-side method is holding the roller drum approximately 6 inches (150 millimeters) away from the longitudinal joint on the first pass toward the paver. This leaves a narrow strip of uncompacted mix. As the roller reverses direction, the roller drum is moved over to overlap 6 inches (150 millimeters) on the cold mat, trapping the narrow strip under the drum and forcing it into place.

For lifts greater than 3 inches (76 millimeters), the rolling process should start 12–15 inches (300–375 millimeters) from the lower unsupported edge, with subsequent passes covering the interior portion of the mat. Succeeding passes of the roller should then gradually progress toward the edges of the spread. The uncompacted edge provides initial confinement during the first pass, thus minimizing lateral movement of the mix. After the central portion of the spread has been compacted, the mix will have cooled slightly and can support the roller. This will allow the edge to be compacted with minimal lateral movement.

Cold-side rolling. A method that is sometimes used is to place the roller on the previously placed lane so that not more than 6 inches (150 millimeters) rides on the edge of the newly spread material. Here it is operated to pinch and press the uncompacted material into place. The roller should continue to roll along this line, its position being shifted gradually across the joint until a thoroughly compacted, neat joint is obtained. Vibration should not be used when the roller is operating on a cold mat. This is normally the least efficient rolling method because the hot mat is cooling while the roller is mostly operating on the cold mat.

Rolling Edges

When rolling a longitudinal joint or HMA edge against an existing concrete gutter or curb lip, the roller should not span the joint in a manner that puts part of the roller weight on the gutter. By bridging the roller on the PCC curb, insufficient compaction along the HMA edge may result. One common method to obtain good HMA compaction along an existing curb is to require the edge of the finished pavement to have an elevation slightly higher than the adjacent gutter. This ensures that the full compactive force of the roller will be on the HMA.

9.5 Troubleshooting

This section is not meant to be an exhaustive summary of all issues relative to compaction. Rather, it will cover the following problem areas: pickup, low density, erratic density, and roller marks. A key to successful compaction is the use of an experienced quality-control technician with a density gauge and infrared thermometer. With these two tools and the authority to work with the roller operators to adjust the rolling pattern as needed, a quality-control technician is a major asset to a paving operation.

9.5.1 Pickup

Mix pickup leaves scars on the mat and is not aesthetically pleasing. *Pickup* is defined as mix sticking to the drums or tires of the roller.

- Pick up on steel drums is typically due to a lack of water-spray systems and scrapers/pads on the equipment.
- Pick up on pneumatic-tire rollers is typically caused by tires that are too cold, are not properly inflated, or lack pads/scrapers.

9.5.2 Low Density

In brief, if the target density is not being achieved, the operation needs to be adjusted. Merely speeding up the rollers is counterproductive and will actually lower density. In fact, slowing the paving and rolling operation should be among the first things to consider, along with the following:

- Establish a new test strip to evaluate the paving-and-rolling process.
- Check the temperature of the mix upon delivery and ensure that it is adequate to achieve density and uniformity between loads.
- Check the calibration of the density gauge against a core to ensure that the gauge is properly calibrated.
- Make sure the paver has not increased speed because the rollers will speed up to catch up.
- Ensure that the rollers have been and continue to be compacting using the established rolling pattern at the proper speed, frequency, and amplitude.
- Check the environmental conditions to determine whether something significant has changed (air temperature, cloud cover, wind speed, shade) because this can affect the time available for compaction.
- Consider shortening the rolling zone.
- Increase the number of coverages and recheck density.

9.5.3 Erratic Density

Erratic density is more difficult to troubleshoot because, by its nature, it is not uniform. In addition to ensuring all the items listed in the previous section have been checked, one should look for changing conditions and erratic roller operation. Changes in density, temperature, mix characteristics, paver speed, etc. should be monitored.

9.5.4 Roller Marks

Roller marks are any marks or deficiencies caused by the rollers.

- Pneumatic-tire rollers can "pick up" if the tires are not hot, properly inflated, and clean.
- Rollers sitting behind a stopped paver will leave marks in the mat that will be almost impossible to remove. If a roller has to stop, it must stop in an area where the mix has cooled.
- If a roller is turned too sharply while it is moving, the edge of the drum can cut or tear the mat.
- If marks are still present after the finish roller has completed, the finish roller is either too light or the time between intermediate and finish rolling was too great (mix cooled too much).
- Poorly maintained equipment, such as worn or pitted drums, can imprint patterns in the mat.

The key to uniform density throughout the compaction operation is keeping track of the little details and monitoring conditions. Compacting hot mix asphalt is not difficult; what is challenging is doing it consistently all the time.

Resources

1. Asphalt Institute. MS-5, *Introduction to Asphalt*. Lexington, KY.
2. Asphalt Institute. MS-22, *Construction of Hot-Mix Asphalt Pavements*. Lexington, KY.
3. The National Asphalt Paving Association. *Hot-Mix Asphalt Paving Handbook 2000*.
4. Roberts, Freddy L., Prithvi S. Kandhal, E. Ray Brown, Dah-Yin Lee, and Thomas Kennedy. *Hot Mix Asphalt Materials, Mixture Design and Construction*. 2nd ed. NAPA Education Foundation, Lanham, MD.

CHAPTER 10

QUALITY CONTROL AND ACCEPTANCE OF HOT MIX ASPHALT

Quality control is the set of actions and considerations necessary to assess and adjust production and construction processes in order to control the level of quality being produced. Quality control practices are usually provided by contractors and/or materials suppliers to monitor the manufacturing process.

Acceptance is the sampling and testing and inspection to determine the degree of compliance with contract requirements. Acceptance practices are usually provided by the owner to determine whether specific terms of the plans or specifications have been met.

Section 10.1 General
10.1.1 Quality-Assurance Programs
10.1.2 Job-Mix Formula

Section 10.2 Sampling Methods
10.2.1 Representative Sampling
10.2.2 Random Sampling
10.2.3 Aggregate Sampling Methods
10.2.4 HMA Mixture Sampling
10.2.5 Asphalt Binder Sampling

Section 10.3 Testing Methods
10.3.1 Aggregate Gradation
10.3.2 Fractured Faces and Aggregate Angularity
10.3.3 Asphalt Content
10.3.4 Maximum Theoretical Specific Gravity
10.3.5 Bulk Specific Gravity (Laboratory Density)
10.3.6 Air Voids
10.3.7 Temperature

Section 10.4 Sampling and Testing Plan for Quality Control
10.4.1 Selection of Sampling Methods and Sample Frequency

10.4.2 Selection of Testing Methods
10.4.3 Written Quality-Control Plan
10.4.4 Test Strip

Section 10.5 Plant Production Quality-Control Data Analysis
10.5.1 Job-Mix Formula Verification
10.5.2 Mixture Control
10.5.3 Plant Adjustments

Section 10.6 Placement Quality-Control Data Analysis
10.6.1 Temperature
10.6.2 Compaction Control Methods
10.6.3 Mixture Compactability
10.6.4 Thickness and Spread Rates
10.6.5 Pavement Smoothness

Section 10.7 Acceptance Point and Test Results
10.7.1 Asphalt Content and Gradation
10.7.2 Aggregate Angularity (Fractured Faces)
10.7.3 Volumetric Properties (Voids, VMA, VFA)
10.7.4 Compaction
10.7.5 Thickness
10.7.6 Ride Quality

Section 10.8 Acceptance Criteria
10.8.1 Historical Acceptance
10.8.2 Modern Acceptance (with Volumetrics)
10.8.3 General Compaction Requirements
10.8.4 Statistical Acceptance—Percent Within Limits (PWL) Process

Section 10.9 Performance-Based Specifications
10.9.1 Specification Definitions
10.9.2 Performance-Related Specifications
10.9.3 Performance-Based Specifications

Section 10.10 Performance Specifications—Warranties
10.10.1 Performance Characteristics
10.10.2 Monitoring Performance Parameters
10.10.3 Dispute Resolution

10.1 General

Long-term performance of hot mix asphalt (HMA) pavements is achieved by using quality materials and placement techniques during construction. The life of a pavement and the cost of maintenance can be very sensitive to seemingly minor variations in materials properties, such as aggregate gradation, asphalt content, mixture volumetrics, and compaction.

Highway agencies and other owners of asphalt pavements use physical test results to evaluate the quality of materials and the methods used in their projects. Each agency's overall quality-assurance program determines testing requirements. These programs generally include three types of tests: quality-control, acceptance, and independent assurance.

10.1.1 Quality-Assurance Programs

The definitions used in this chapter have been adapted from the TRB Transportation Circular, *Glossary of Highway Quality Assurance Terms.* The following terms are presented relative to quality-assurance programs. This chapter focuses mainly on these terms as they relate to sampling and testing, and is limited to aggregates and HMA mixtures.

Quality Assurance. All planned and systematic actions necessary to provide confidence that a product or facility will perform satisfactorily in service.

Quality Control (also called Process Control). Actions and considerations necessary to assess and adjust production and construction processes in order to control the level of quality being produced in the product. Quality-control practices are usually provided by contractors and/or materials suppliers to monitor the product manufacturing process.

Acceptance. Sampling and testing, or inspection, to determine the degree of compliance with contract requirements. Acceptance practices are usually provided by the owner or his designated representative to determine whether specific terms of the plans and specifications have been met.

Independent Assurance. Management tool that requires a third party, not directly responsible for process control or acceptance, to provide an independent assessment of the product and/or the reliability of test results obtained from the process-control and acceptance testing. This third-party testing and inspection is usually provided by the owner but accomplished by a laboratory that is not involved in the quality-control or acceptance.

Table 10.1 is a general outline of the tests that are typically included in a quality-assurance program for HMA construction. Regular sampling and

Table 10.1
Testing Outline for Quality-Assurance Programs

I. Preproduction Quality-Control Sampling and Testing (Supplier Provided)
A. Continuous aggregate gradation during crushing
B. Periodic aggregate quality during crushing as needed
C. Aggregate Sampling for mix design (must be representative)
D. Asphalt cement samples for mix design
E. Mineral filler and any other additives for mix design

II. Job-Mix Formula Approval and Verification (Supplier Provided)
A. Aggregate gradation
B. Aggregate physical properties as required
C. Air voids, VMA, VFA
D. Stability and strength testing as required
E. Moisture susceptibility testing
F. Asphalt content calibration (for ignition or nuclear methods)

III. Quality-Control Testing During Mixture Production (Supplier Provided)
A. Maximum theoretical specific gravity
B. Bulk specific gravity
C. Air voids
D. Aggregate gradation
E. Asphalt content

IV. Quality-Control Testing During Paving (Provided by Paving Contractor)
A. Thickness and/or spread rates
B. Compaction
C. Smoothness

V. Acceptance Testing (Provided by Owner/Agency)
A. Asphalt content
B. Aggregate gradation
C. Air voids, VMA, VFA
D. Compaction
E. Thickness or spread rate
F. Smoothness (check for bumps)
G. Roughness (Ride Quality)

VI. Independent Assurance and/or Correlation Testing (Provided by Owner/Agency)
A. Asphalt content
B. Aggregate gradation
C. Air voids, VMA, VFA
D. Compaction
E. Thickness or spread rate
F. Smoothness (check for bumps)
G. Roughness (Ride Quality)

testing of the asphalt binder materials delivered to the project are equally, if not more, important and should be part of the agency quality-assurance program as well. See Chapter 2, "Petroleum Asphalt," for more information on asphalt cement and its test methods.

Most agency quality-assurance programs include quality-control testing and acceptance testing on all their projects. The use of independent assurance concepts on nonfederal projects, however, is increasing, especially for projects using multiple funding sources and for alternative contracting methods such as design-build and public-private partnerships. State and local agencies in the United States are required to implement quality

assurance programs with independent assurance testing for projects that are funded by the federal government.

Implementation of testing programs varies from one agency to another. Some agencies require contractors to provide specific quality-control testing, while others rely only on their own test results. Some agencies are now using contractor quality-control test results in the acceptance decision. When alternative contracting mechanisms are used by public agencies, such as "Design Build" and "Construction Manager at Risk," new paradigms may be needed to understand each entity's responsibility in the quality-assurance program.

Quality-assurance programs lead to comparisons of test results from multiple laboratories. This has created an increased emphasis on accuracy and repeatability of test results. Many agencies now require laboratories and technicians to be pre-qualified before providing testing in their quality-assurance programs. Many state highway agencies have their own certification programs for laboratories and technicians, but national certifications are also available from organizations such as the American Concrete Institute (ACI), the National Institute for Certification of Engineering Technicians (NICET) and the Asphalt Institute Mix Design Technologies Certification Program for technicians, and the AASHTO Materials Reference Laboratory (AMRL), or the National Voluntary Laboratory Accreditation Program (NVLAP) for laboratories. ASTM D 3666, "Minimum Requirements for Agencies Testing and Inspecting Bituminous Paving Materials," is a standard specification to address the testing laboratory and technician capabilities in performing tests on bituminous materials. This standard can be a good reference when establishing a quality-assurance program.

10.1.2 Job-Mix Formula

For quality-control and acceptance of hot mix asphalt, specification limits are often based on a job-mix formula (JMF) established as part of the mixture design. The mix design will produce a JMF that establishes the target values for production of the HMA for the project. The JMF is the combination of aggregate and asphalt binder materials proposed for use on a project that, when tested using the required mix design procedures, yields results that meet all the established design criteria. At a minimum, the JMF should include the following target parameters from the mix design:

- Asphalt content
- Aggregate gradation
- Laboratory air voids
- Stockpile blend percentages
- Hot-bin blend percentages (for batch plants)
- Other parameters specified for quality-control such as mixture volumetrics or aggregate angularity
- Specific materials used (that is, sources for asphalt cement, admixtures, aggregate descriptions, etc.)

The JMF that results from the mix design becomes the "recipe" for producing that mixture in the hot plant. Specifications often dictate how closely the HMA supplier must adhere to this recipe.

The specifications may require, for example, that the binder content of the material produced shall not vary by more than 0.4 percent from the binder content in the JMF (established in the approved mix design). The concept is that the mix design provides evidence that a mixture made with given aggregates, at a certain binder content and gradation, and compacted with a specific amount of energy will comply with specific quality indicator tests. These indicators can include parameters such as aggregate absorption, tensile strength ratio (TSR), volumetric properties, and strength parameters such as Marshall stability. The premise is that if the mixture is controlled to be reasonably compliant with the fundamental mix-design properties (that is, gradation, binder content, and air voids), then the mixture produced will essentially comply with the quality properties (TSR, Stability, and VMA) established in the mix design. This assumption allows fewer tests for quality-control and acceptance than were required in the mix design. The assumption also requires a revised mix design if any of the quality-control or acceptance parameters change significantly.

More information on mix designs and their procedures can be found in Chapter 4, Hot Mix Asphalt (HMA) Design or in Asphalt Institute's publications MS-2, *Mix Design Methods for HMA and Other Hot-Mix Types* or SP-2, *Superpave Mix Design*.

10.2 Sampling Methods

The methods used for sampling HMA can have a significant effect on the outcome of the test results. It is critical that good sampling methods are used and that the specified procedures are carefully followed to get an accurate representation of the materials being produced.

10.2.1 Representative Sampling

The primary goal is to obtain samples that are representative of the products being evaluated. Since test results are used to adjust asphalt mixing plants, it would be counterproductive to overall quality to obtain a sample from a particularly "bad" or a particularly "good" portion of the work. Consider as an example an asphalt plant quality-control sample taken from a nonrepresentative portion of the asphalt mixture with an unusually low binder content. The proper response from the plant operator to a low asphalt content test result would likely be to increase the amount of asphalt binder being incorporated into the mix. After such an adjustment, he would inadvertently have the asphalt plant set too high for asphalt binder content.

Representative samples of the average production are critical for the effective use of quality-control and acceptance samples. Inspection is more appropriate for identifying non-uniform portions of the work. Once inspection has identified a portion of the work that is non-uniform (different from the rest of the work), or for which defective materials are suspected, additional samples representing that specific element of the work should be taken. Test results from such samples should never be used as part of the lot averages, and certainly not for lot acceptance. They only represent the specific problem areas identified by inspection.

As an example, consider an asphalt pavement that contains a 50-foot (15.2 meters) length of roadway that is littered with manholes and valve covers that do not allow the regular compaction equipment into that isolated area. By visually observing the entire pavement, an inspector or technician would be able to recognize that the isolated area will not have density values that are representative of the pavement compaction for that day's placement. Density tests taken there would represent that area only.

Regardless of the methods being employed, good samples will be obtained if a common-sense approach is combined with proper sampling technique. Visual observations of the entire element to be sampled can be valuable in selecting sample locations or number of sample collection points, and in verifying that representative samples can be retrieved.

It is also important to get a sufficient quantity of material when sampling. The sample size is important not only to obtain enough material to complete the tests but also to achieve a representative sample. As an exaggerated example, consider a graded gravel material that contains 15 percent of rock that is retained on the ¾-inch (19 millimeters) screen and passes the 1-inch (25 millimeters) screen. Each of these +¾-inch (19 millimeters) rocks weigh about 250 grams. Assume that a sample size of only 500 grams is obtained. If the sample taken happens to contain one large rock, then the gradation test will yield 50 percent retained on the ¾-inch (19 millimeters) sieve. If the 500-gram sample scoop does not happen to capture a large rock, then the gradation test would yield 0 percent of that size. In reality, the gradation is somewhere in between 0 percent and 50 percent. As the sample size increases, the test result is no longer impacted by the size of the sample—it is only impacted by the true gradation of the materials being tested. This is the point that establishes minimum sample sizes. Typical minimum sample sizes for aggregates and for HMA are presented in Table 10.2.

Production Lots

The quantity of HMA produced on a given project is usually divided into defined portions called "lots." Quality assurance and quality-control plans usually require samples to be taken at defined intervals within these lots. The intervals are generally measured in any of the following two ways.

1. **Time intervals.** Typical lot sizes when sampling by time interval would include either a half day's production, or a full day's production.
2. **Quantity intervals.** Typical lot sizes when sampling by quantity interval would include 5,000 or 10,000 tons (4,536 to 9,071 tonnes), depending on anticipated production rates. Lots based on linear feet (meters) of paving have also been used.

Table 10.2
Minimum Sample Size for Aggregates

Nominal Maximum Aggregate Size		Aggregate Minimum Sample Weight		HMA Minimum Sample Weight	
English	Metric	English	Metric	English	Metric
No. 8	2.36 mm	10 lb	5 kg	4 lb	2 kg
No. 4	4.75 mm	10 lb	5 kg	4 lb	2 kg
⅜ in	9.5 mm	10 lb	5 kg	8 lb	4 kg
½ in	12.5 mm	20 lb	10 kg	12 lb	5 kg
¾ in	19.0 mm	30 lb	20 kg	16 lb	7 kg
1 in	25.0 mm	50 lb	30 kg	20 lb	9 kg
1½ in	37.5 mm	70 lb	50 kg	25 lb	11 kg

Lot sizes are normally established by project specifications. It is important to have lot sizes defined in advance of construction. The size and method of measurement of a lot will be used to determine locations and quantities of quality-control and quality assurance tests. The lot will also be used to define what portion of the work is represented by a given set of test results. The limits of the lot are critical if corrective action or removal is required.

10.2.2 Random Sampling

Use of random sampling techniques is an important part of obtaining unbiased representative samples. Random sampling methods can be simple to use if good forms and/or other tools are used. With widespread availability of personal computers, simple spreadsheet documents can be used to easily generate sample locations and make implementation less daunting.

Random sampling can be applied to any of the sampling interval methods (time, quantity, and length). Random sampling is achieved by taking the number of samples to be taken that day and distributing the times, quantities, or locations selected for each sample in a random way. For example, assume that the production of 5,000 tons (4,536 tonnes) of material is planned and a sample frequency for every 1,000 tons (907 tonnes) has been selected. Nonrandom techniques may lead planners to take the first sample at about 1,000 tons (907 tonnes) and the second sample at about 2,000 tons (1,814 tonnes) and so on until all five of the samples for that day are collected. Using random sampling, a random number (numbers between 0 and 1 obtained from a chart) would be assigned to each of the five samples. Each of the random numbers is multiplied by the total tonnage (5,000 or 4,536) to determine at what point in production to take the sample. For example, if the random number for one of the samples is 0.43218, then that sample would be taken when the day's accumulative tonnage reaches 2,160 tons or 1,959 tons ($5{,}000 \times 0.43218 = 2{,}160$) ($4{,}536 = 0.43218 \times 1{,}960$).

An accurate estimate of the day's production is needed to determine random sample locations. In some cases, lots with different sizes may have a different number of tests. For example, if the project estimate for production is 6,520 feet (1,978 meters), and the quality-control plan establishes the sample frequency for coring at every 1,000 feet (305 meters), then that lot will contain seven core locations.

Table 10.3 shows how to calculate the stations for random locations for cores. Another set of random numbers would need to be used to determine how far right of the centerline the location would be. So, as an example, the final location of a given core location could be "Station 15 + 83 (0 + 482 meters), and 7 feet (2.1 meters) right of centerline."

Careful review of Table 10.3 reveals that the sample locations are not very evenly spaced throughout the length of the lot. Samples 6 and 7, for example, are only 14 feet (4.3 meters) away from each other. This type of occurrence is a natural consequence of using random numbers.

Table 10.3
Sample Random Number Sample Locations

Sample	Random Number (from chart)	Total Length of Lot in feet (m)	Distance Ahead in feet (m)	Station, feet (m)
Begin Lot				12+00 (0+365)
1	.05874	6520 (1905)	383 (117)	15+83 (0+482)
2	.07086	6520 (1905)	462 (141)	16+62 (0+506)
3	.18591	6520 (1905)	1212 (369)	24+12 (0+735)
4	.28612	6520 (1905)	1866 (569)	30+66 (0+934)
5	.69055	6520 (1905)	4502 (1373)	57+02 (1+738)
6	.84524	6520 (1905)	5511 (1679)	67+11 (2+044)
7	.84733	6520 (1905)	5525 (1684)	67+25 (2+049)
End Lot				77+20 (2+353)

The use of sublots, however, will spread the samples more evenly across the length of the lot, while still maintaining random sample locations. Many agencies use sublots, or some other form of stratified random sampling in establishing random locations. The same example is presented in Table 10.4, except that the locations are stratified throughout the length of the lot in equal increments. The stratified random sampling method is recommended unless project specifications stipulate another.

An example of a spreadsheet solution for selecting random locations for cores is shown in Figure 10.1A and 10.1B.

The above examples present sampling based on length (or station) increments. They are useful if samples are to be taken from the roadway. These same methods and spreadsheet solutions could be applied if samples were to be obtained based on accumulated tonnage. Sampling increments based on tons of HMA produced would be used for samples to be taken at the plant.

10.2.3 Aggregate Sampling Methods

The various methods used to obtain aggregate samples are discussed briefly in the following paragraphs. Sampling procedures in AASHTO T 2 or ASTM D 75 should be carefully followed when sampling aggregates.

Stockpile Sampling

Stockpile sampling techniques are used for obtaining test samples for the various aggregates that are incorporated into the mix (Figure 10.2). Stockpiles of the individual aggregates may be sampled during the crushing and screening operation in order to facilitate more consistent and better known

Table 10.4
Sample Stratified Random Number Sample Locations

Sample	Distance, feet (m)	Cumulative Distance, feet (m)	Random Number	Distance Ahead, feet (m)	Cumulative Distance Ahead, feet (m)	Station, feet (m)
Begin						12+00 (3+65)
1	931 (284)		.05874	55 (17)	55 (17)	12+55 (0+382)
2	931 (284)	931 (284)	.07086	66 (20)	997 (304)	21+97 (0+685)
3	931 (284)	1862 (567)	.18591	173 (53)	2035 (620)	32+35 (1+305)
4	931 (284)	2793 (851)	.28612	266 (81)	3059 (932)	42+59 (2+237)
5	931 (284)	3742 (1135)	.69055	643 (195)	4385 (1336)	55+85 (1+702)
6	931 (284)	4655 (1419)	.84524	787 (240)	5442 (1658)	66+42 (2+024)
7	934 (284)	5586 (1703)	.84733	791 (241)	6377 (1943)	75+77 (2+309)
End		6520 (1987)				77+20 (2+353)

gradations of the constituent materials to be incorporated into the mix. Stockpile samples are often used for development of mixture designs, so it is critical that the sampling techniques produce truly representative samples.

Because stockpiled aggregates have a tendency to become segregated, collecting representative samples from stockpiles can be challenging. Visual observation of the pile, some knowledge about its production, a good common-sense approach, and proper sampling techniques are required to overcome the challenges.

Conveyor Belt Sampling

Conveyor belt sampling techniques are generally used to obtain aggregate samples (Figure 10.3). They can be used to sample individual aggregate materials in production, as they are being transported to a stockpile, or combined aggregates that have been blended and are being transported to the dryer of a hot plant. They generally provide good samples if the belt is properly cut and cleaned, but the procedures are inconvenient because the belt conveyor must be stopped to retrieve the sample.

Some plants are equipped with automatic sampling devices that can retrieve samples without stopping conveyor belts or other plant production equipment. These automatic samplers are used to collect aggregate or hot mix samples. Such devices can be very useful, but should be used

Lot 2

Using Spreadsheet Random Number Generator
This Spreadsheet Develops Stratified Random Locations

Begin Station	**114.00**		*Date of Coring*	**12/13/2006**
End Station	**132.50**		*Date asphalt placed*	**12/12/2006**
Asphalt Placed	**1850**	*Feet*		
*Number of Cores**	**4**	(Adjust number of cores to achieve required sample frequency)		
Width of Mat	**10**	*Feet*	(Placement width)	

Selection of Station—Reference from Beginning Station

Core Number	Distance* Length (ft)	Cumulative Distance	Random Number	Distance Ahead	Cumulative Distance Ahead	Station
Begin						**114.00**
1	463	463	0.61961	287	287	**116.87**
2	463	925	0.58878	272	735	**121.35**
3	463	1388	0.38870	180	1105	**125.05**
4	463	1850	0.74027	342	1730	**131.30**
End						**132.50**

*
Note: – This distance is the length paved divided by the number of cores.
– Adjust the number of cores to satisfy project sample frequency requirements.

Selection of Offset—Distance from Left Edge

Core Number	Distance Width (ft)**	Random Numbers	Distance from Left Edge
1	8	0.90142	**8**
2	8	0.73757	**7**
3	8	0.32955	**4**
4	8	0.90328	**8**

**
Note: – This distance is 2 feet less than the width of the paving pass.
– The last foot on the right and left edges are not sampled as part of the mat.
– Separate sampling to represent edge or joint densities could be done to represent that portion of the pavement.

Figure 10.1A Random Core Locations Worksheet (English)

Lot 2

Using Spreadsheet Random Number Generator
This Spreadsheet Develops Stratified Random Locations

Begin Station	**3 + 474**		*Date of Coring*	**12/13/2006**
End Station	**4 + 038**		*Date asphalt placed*	**12/12/2006**
Asphalt Placed	**564**	*Meters*		
*Number of Cores**	**4**	(Adjust number of cores to achieve required sample frequency)		
Width of Mat	**3**	*Meters*	(Placement width)	

Selection of Station—Reference from Beginning Station

Core Number	Distance* Length (m)	Cumulative Distance	Random Number	Distance Ahead	Cumulative Distance Ahead (m)	Station
Begin						**3 + 474**
1	141	141	0.61961	87.4	141.0	**3 + 615**
2	141	282	0.58878	83.0	224.4	**3 + 699**
3	141	423	0.38870	54.8	336.8	**3 + 811**
4	141	564	0.74027	104.4	527.4	**4 + 001**
End						**4 + 038**

*
Note:
- This distance is the length paved divided by the number of cores.
- Adjust the number of cores to satisfy project sample frequency requirements.

Selection of Offset—Distance from Left Edge

Core Number	Distance Width (m)**	Random Numbers	Distance from Left Edge
1	2.44	0.90142	**2.44**
2	2.44	0.73757	**2.13**
3	2.44	0.32955	**1.22**
4	2.44	0.90328	**2.44**

**
Note:
- This distance is .61 meter less than the width of the paving pass.
- The last 0.3 meter on the right and left edges are not sampled as part of the mat.
- Separate sampling to represent edge or joint densities could be done to represent that portion of the pavement.

Figure 10.1B Random Core Locations Worksheet (Metric)

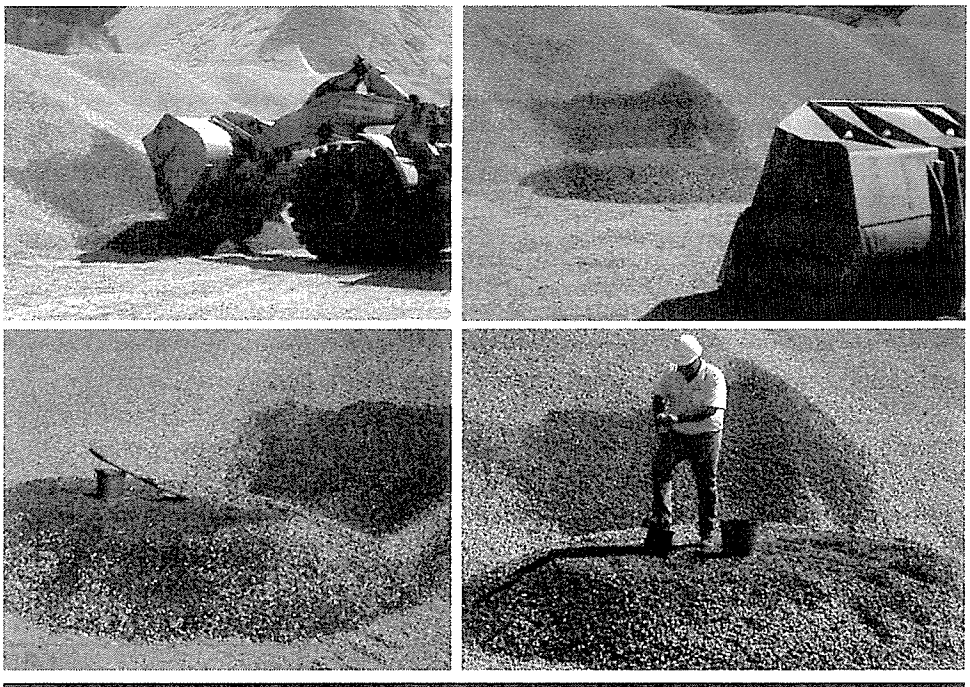


Figure 10.2 Stockpile Sampling

only after sufficient duplicate samples have been taken to compare them to belt samples in order to verify that the sampling device produces representative samples. It is generally necessary to verify the sampling devices throughout a range of belt speeds or production rates.

Hot Bin Sampling

For batch plants, aggregate samples for quality-control are often taken from the hot bins (Figure 10.4). Proper sampling devices that cut through the entire discharge stream are needed to achieve representative samples.

10.2.4 HMA Mixture Sampling

Sampling methods are available for sampling HMA at many points along the construction process, including at the plant, from delivery trucks, from windrows at the job site, and from the paving grade behind the paving machine. For all these sampling methods, the procedures described in AASHTO T 168 or ASTM D 979 should be followed. Samples of the compacted HMA materials can also be taken by cutting cores from the cooled pavement following methods described in ASTM D 5361 or AASHTO T 168. A brief discussion of each sampling location is presented below.

Plant Sampling

Some asphalt mixing plants are equipped with sampling devices to obtain samples before delivering to the haul units. Such devices can be



Figure 10.3 Belt Sampling

very useful but should be used only after sufficient duplicate samples have been taken to compare them with belt cuts or other samples taken according to ASTM or AASHTO standard methods in order to verify the sampling device produces representative samples. It is generally necessary to verify the sampling devices over a range of production rates.

Truck Sampling

Truck sampling (Figure 10.5) techniques are used to obtain samples of the HMA. Quite often, the samples are taken immediately after loading from the plant where sampling platforms allow the technicians to stand at the height of the truck bed. HMA in the back of the trucks can become segregated, making it more difficult to obtain representative samples. Good adherence to the specified sampling methods is required.

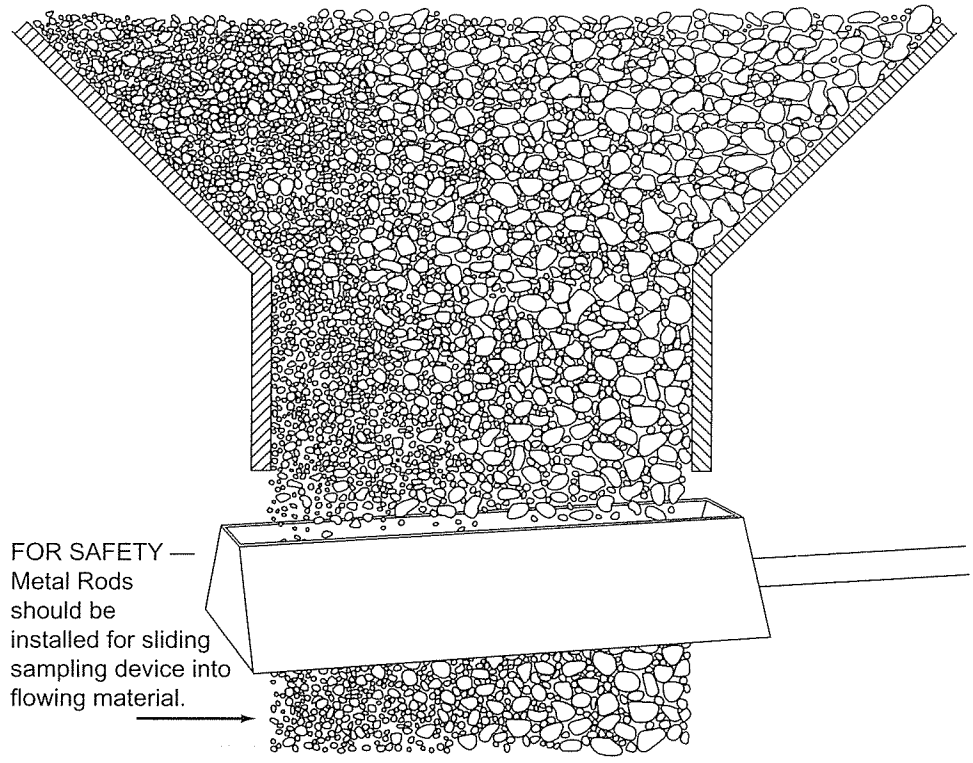


Figure 10.4 Hot Bin Sampling

Windrow Sampling

Windrow sampling (Figure 10.6) techniques are generally used for aggregates, but they have also been used to obtain samples of HMA. Windrow samples can be used if "belly dump" haul units are used. These units will create a windrow of HMA that can be sampled before placement into the paving machine. The HMA can become segregated in the windrow, so strict adherence to the sampling procedure is necessary to obtain representative samples.

In-Place Pavement Sampling

Representative portions of HMA can be obtained by sampling behind the paving machine. These samples are taken immediately after placement by the paving machine but before any rolling. The use of plates (Figure 10.7) placed on the grade in front of the paver is recommended except when thin lifts are being placed.

Core Sampling

Core sampling is the most common form of sampling from completed HMA pavements (Figure 10.8). Portable electric drills with wet-cut diamond-studded core barrels provide rapid removal with little or no damage to the specimens. These samples are normally used to evaluate compaction and thickness.

Once removed from the pavement, the test specimens must be handled with care because they are very sensitive to damage from heat and point loads.

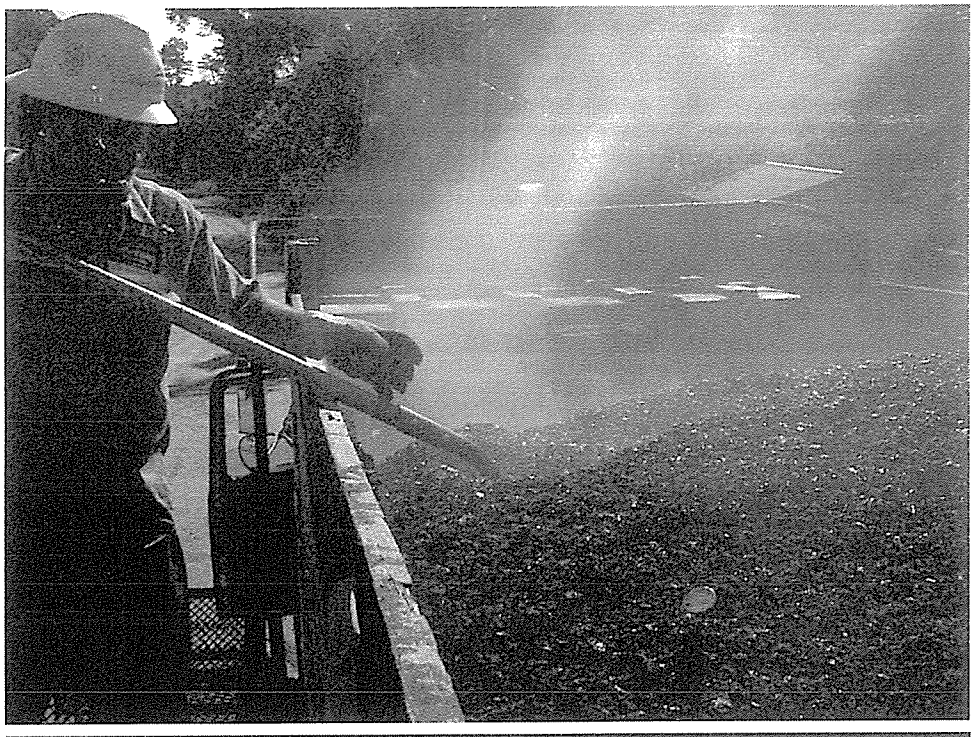


Figure 10.5 Truck Sampling

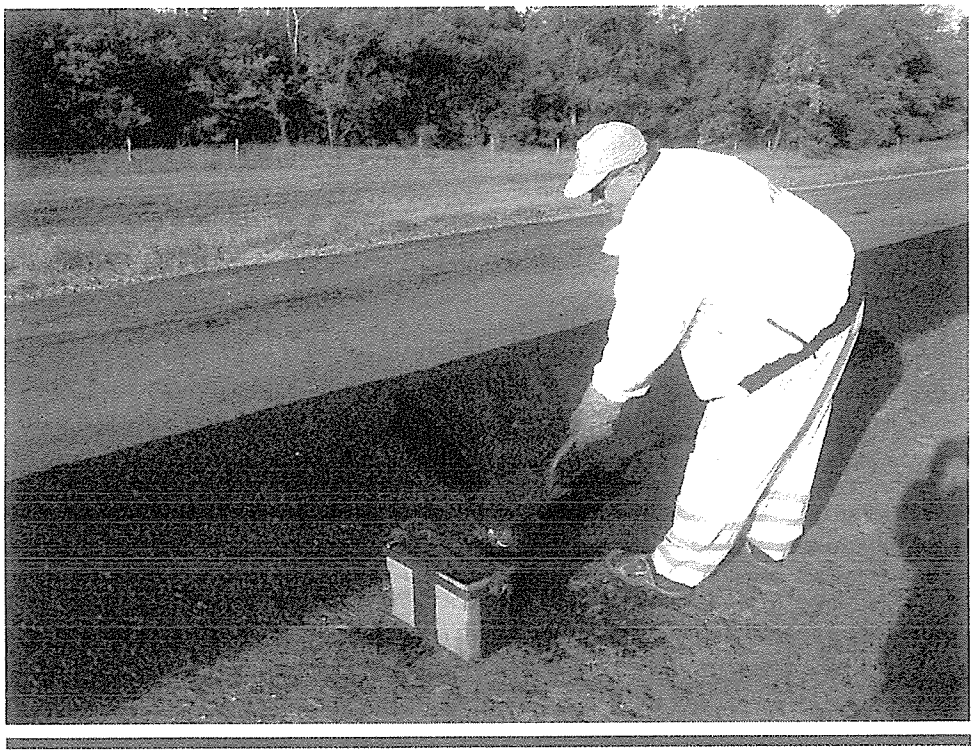


Figure 10.6 Windrow Sampling



Figure 10.7 Plate Sampling

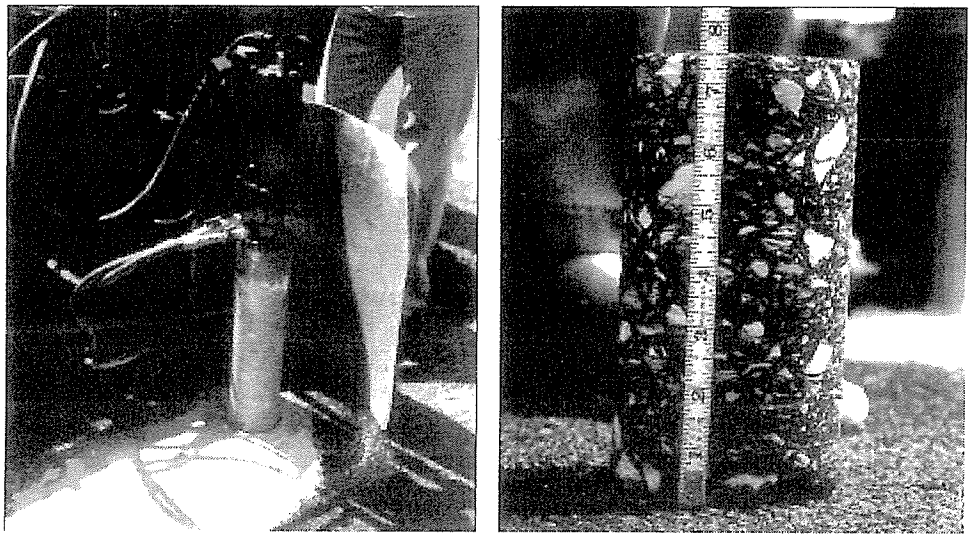


Figure 10.8 Asphalt Coring

10.2.5 Asphalt Binder Sampling

Asphalt binder can be sampled from delivery truck tanks, from the plant's asphalt storage tank, or from delivery lines from the tank to the asphalt mixing plant. Samples from the delivery trucks are more representative of the binder delivered by the supplier, but samples from the storage tank or the delivery lines to the hot plant are more representative of the binder that is incorporated into the mix. The procedure involves opening a valve that is usually located near the bottom of the tank. Sample containers must be new, clean, and dry. At least one quart of asphalt should be allowed to flow out of the valve and discarded before filling the sample container. It should be sealed immediately after sampling, wiped with a clean dry cloth, and properly marked to identify the material for shipping to the laboratory.

Appropriate personal protective equipment should be worn around the hot asphalt. Generally, representatives of the asphalt supplier or HMA producer will obtain the samples because of safety and insurance concerns. The quality-control or acceptance technician then takes immediate possession of the sample.

Procedures identified in AASHTO T 40 or ASTM D 140 should be followed when sampling asphalt binder.

10.3 Testing Methods

The ultimate goal of HMA testing is to accurately measure the appropriate quality of the materials used. Laboratories must strive to reduce the variations caused by testing so that test results can be confidently used to evaluate the materials and production methods. However, all parties involved in quality-control and the acceptance testing process must understand the variability of test results that comes from the testing methods themselves. All parties must recognize that the variability related to the test method can be reduced but never eliminated. This is especially important for test parameters that will be used for pay factor determination, rejection or acceptance of the materials produced.

The test methods used will be established in the quality-assurance plan and the project specifications. Agencies or owners may change or modify the standard test procedures. It is important that the laboratories get a clear understanding of the methods to be used before starting the project. Agencies modify standard methods to reduce variation from one lab to another. ASTM D 979, *Standard Practice for Sampling Bituminous Paving Mixtures,* for example, covers sampling of HMA at points of manufacture, storage, delivery, or in place. A variety of sampling methods are included within this one standard, so an agency's project specifications might stipulate which method is required. Many agencies hold pre-paving meetings for projects. These meetings are a good forum for clarifying the methods and the roles of the parties involved in quality-control and acceptance testing.

The following sections describe testing usually provided for quality-control and acceptance of HMA.

10.3.1 Aggregate Gradation

A tremendous amount of work goes into processing the aggregate materials to achieve a specific gradation in the finished product. The final gradation can be impacted by processes in the mining, crushing, washing, stockpiling, and handling, or in the blending of the aggregates. Quality-control must be exercised throughout all processes in order to achieve the final gradation.

Different sampling methods may be used in the various stages of processing aggregates, but the test procedures are generally the same. An exception to this is the use of unwashed gradations for quality-control testing. Washed gradations are more accurate and are generally used for acceptance tests; however, it is appropriate in some instances to use unwashed

gradations for quality-control purposes. Consider, for example, gradation testing performed for quality-control after crushing. If the materials are crushed in a dry state, and they are separated into stockpiles containing certain gravel sizes, it may be appropriate to test these gravels without washing. A visual observation can verify that little dust remains adhered to the aggregate particles, that the gradation will not likely be affected by more than one percent, and that the quality-control manager can estimate that the percentage passing the No. 200 screen would be slightly higher. Quality-control testing used to evaluate a washing operation's effectiveness at removing fines, on the other hand, may require washed gradations. It is important to remember that this is not the acceptance point. The aggregates still have to be blended and further processed before being incorporated into the HMA mixture, and the quality-control manager must also estimate aggregate degradation during this processing. One can see that the added accuracy gained by washing gravel materials may not benefit controlling the crushing operation. Using non-washed gradations could actually provide more benefit because it gives more timely results and might allow the technician to provide a higher test frequency.

Test procedures for aggregate gradation can be found in AASHTO T 27 and T 11 or ASTM C 117 and C 136.

10.3.2 Fractured Faces and Aggregate Angularity

Aggregate angularity has long been known to affect the void properties and compactability of HMA mixtures. Traditionally, aggregate angularity is tested by physically separating the aggregate particles in a given test sample containing fractured surfaces from those containing no fractured surfaces. Then the two masses are weighed and the percentage of the sample containing fractures is determined as a percentage by weight. Typical ranges of fractured-faces specifications are between 75 percent and 100 percent. Some specifications require a certain percentage of particles with two or more fractures. The fracture faces test can be performed on coarse and fine aggregate, but the fine aggregate testing can be very tedious and is prone to technician error. There is also a practical limit to the smallest particle sizes that can be evaluated.

The difficulty of testing the fractured faces of fine aggregates, combined with the knowledge that the fine-aggregate texture affects the volumetrics of mixtures more than the coarse aggregate texture, led to the development of the uncompacted voids test. In this test, the fine aggregate is poured through a standard funnel into a standard cup. The cup is gently struck off and the weight recorded. One can imagine that a highly angular aggregate would bulk up in the cup and render a lighter weight, whereas a rounded aggregate would pack into the cup and render a heavier weight. The aggregate specific gravity must be known for the aggregates tested,

and a void content as a percent by volume can be calculated to render the "uncompacted voids."

Many quality-control specifications require aggregate-angularity testing. This is because it is necessary to maintain or control the surface texture of the aggregate particles to control the mixture volumetrics. Test procedures for coarse aggregate angularity (CAA) or fractured faces can be found in ASTM D 5821. Test procedures for fine aggregate angularity (FAA) or uncompacted voids can be found in AASHTO T 304 or ASTM D 5821.

10.3.3 Asphalt Content

In the past, testing for asphalt content was done with chlorinated industrial solvents such as trichloroethylene or methylene chloride to wash the asphalt out of an HMA sample. This provided an asphalt cement content and left the washed aggregate for gradation testing. Because of the high cost of purchasing and disposal, and the human health risks to those handling the solvents, the industry is increasingly turning to different methods of testing asphalt content.

One method that has been developed is the nuclear asphalt content gauge. These gauges have a radioactive source and backscatter detectors that are sensitive to the presence of hydrogen ions. Asphalt binder contains a significant number of hydrogen ions and therefore the gauges can be calibrated to provide asphalt content measurements. These gauges are currently used by some agencies as a replacement for solvent extractions. They also have an added benefit for quality-control purposes in that they provide results within minutes of sampling. In order to own and operate these gauges, however, users must have radiation safety licenses with their respective regulatory agencies and the operators must be trained and certified to use the equipment. Recent changes to licensing rules in the United States have added additional security requirements to the already stringent radiation safety rules.

Another recent development for testing asphalt content is the ignition method. This method uses a large muffle furnace to heat the asphalt mixture specimen to temperatures exceeding 1000°F (538°C); at these temperatures, the asphalt cement is completely burned up and all that remains is a light ash. The ash can be easily washed from the remaining aggregate and the aggregate tested for gradation. This method also requires calibration because some of the rock particles can burn at the elevated temperatures. Therefore, calibration factors for asphalt cement content and for gradation must be applied. For most aggregates, the calibration factors are small and manageable, but certain aggregates, especially those containing dolomites and limestone, require special handling and modifications to the calibration and testing methods.

Table 10.5 provides standard test methods that can be used for determining asphalt content of HMA.